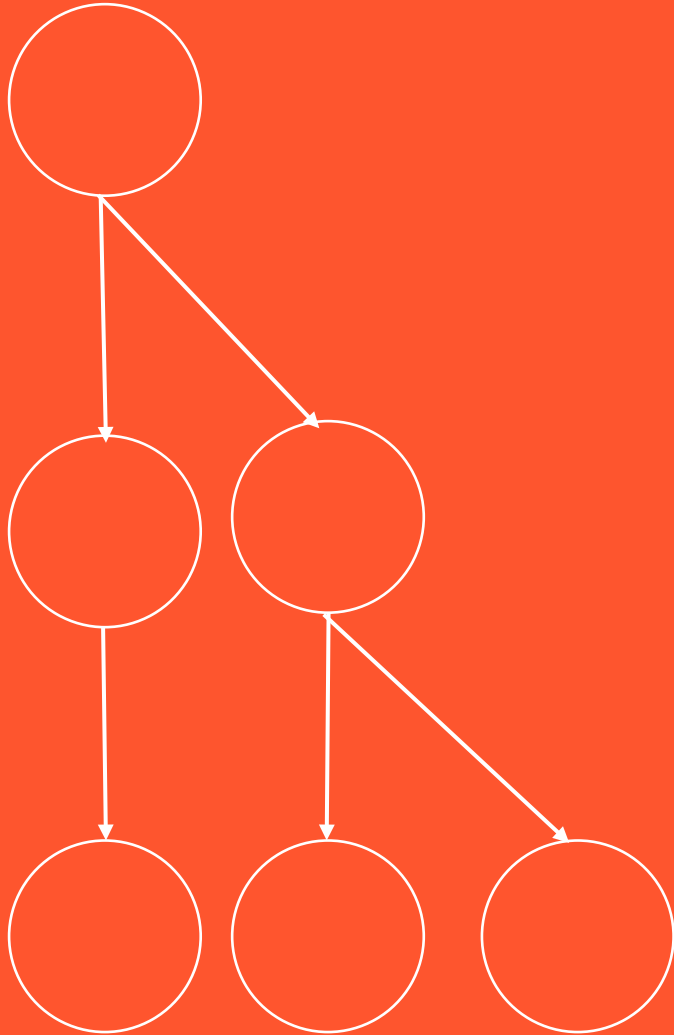


BCOG 100

Brains and Reinforcement Learning

Lecture 1: The Computational Level



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Reading Check 6-1

Password: reward

Question: How do classical and instrumental/operant conditioning relate to hebbian learning?

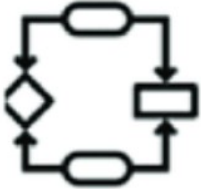


Brains and Reinforcement Learning



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Recap & Overview

From circuits and Hebb to actions over time

Today:

- What problem does reinforcement learning solve?
- Why natural selection favored organisms that could do more than simple associations

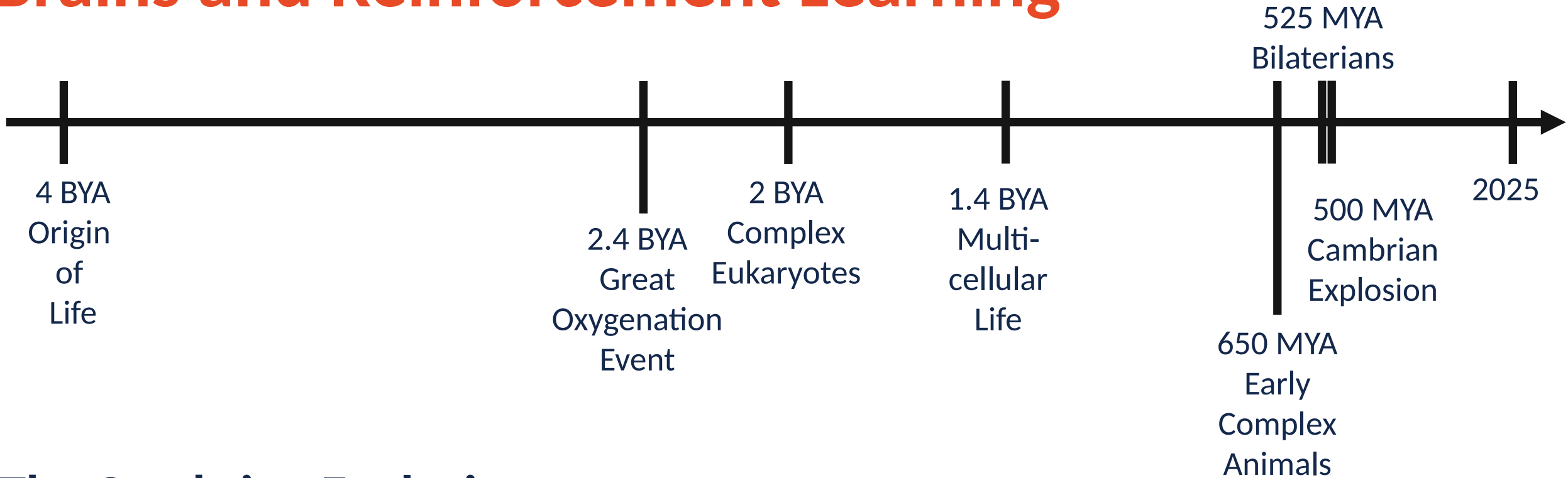
Brains and Reinforcement Learning



Why Hebb and Classical Conditioning Isn't Enough

- Hebbian Learning and Classical Conditioning = co-occurrence learning
- Temporal credit assignment problem

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The Cambrian Explosion

- 540–485 million years ago
- Rapid diversification of animal life
- Predator–prey arms race drives complexity

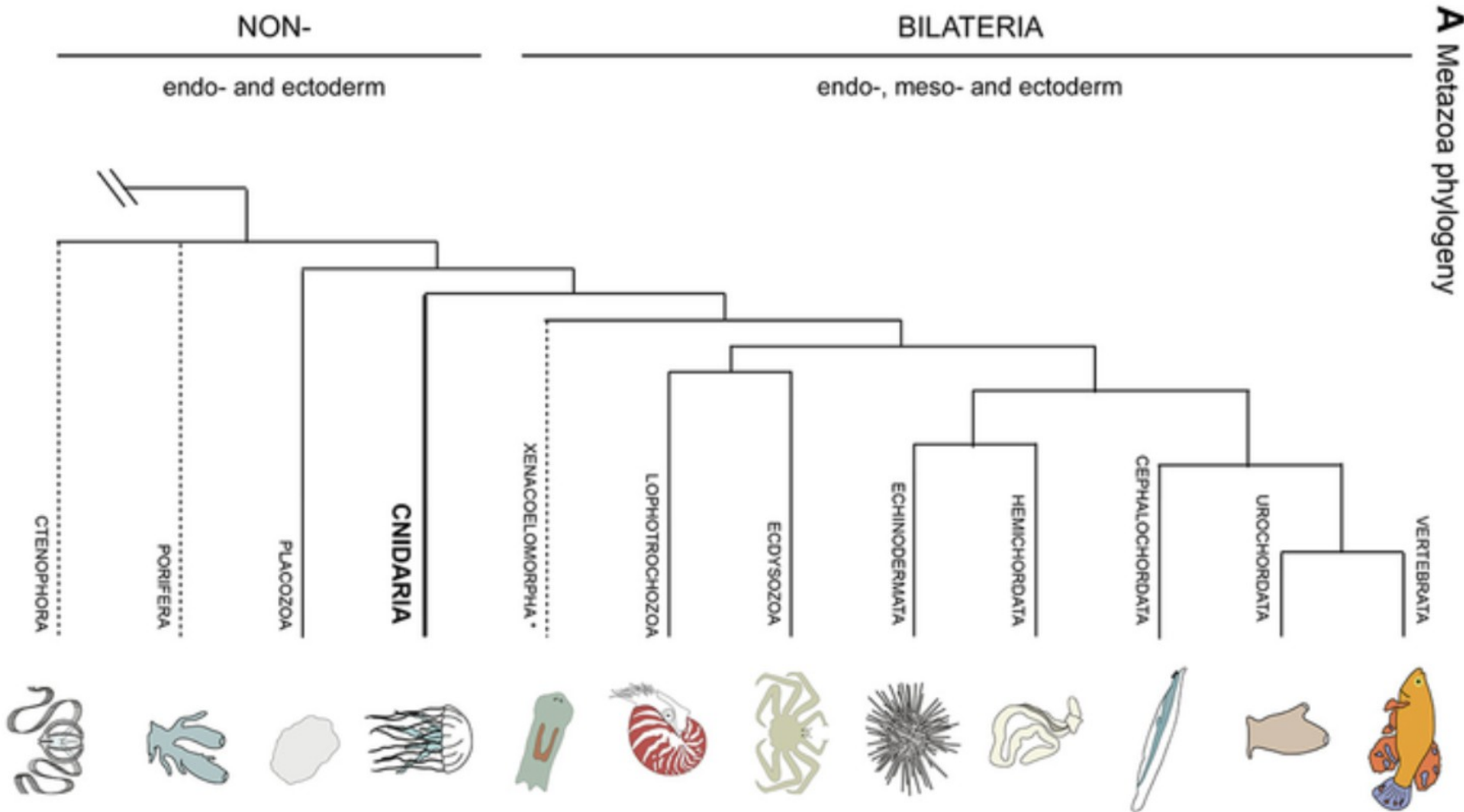
Brains and Reinforcement Learning



Brains Before Reinforcement Learning

- Early bilaterians: simple circuits
- Valence and steering mechanisms
- Reflexes, but limited flexibility

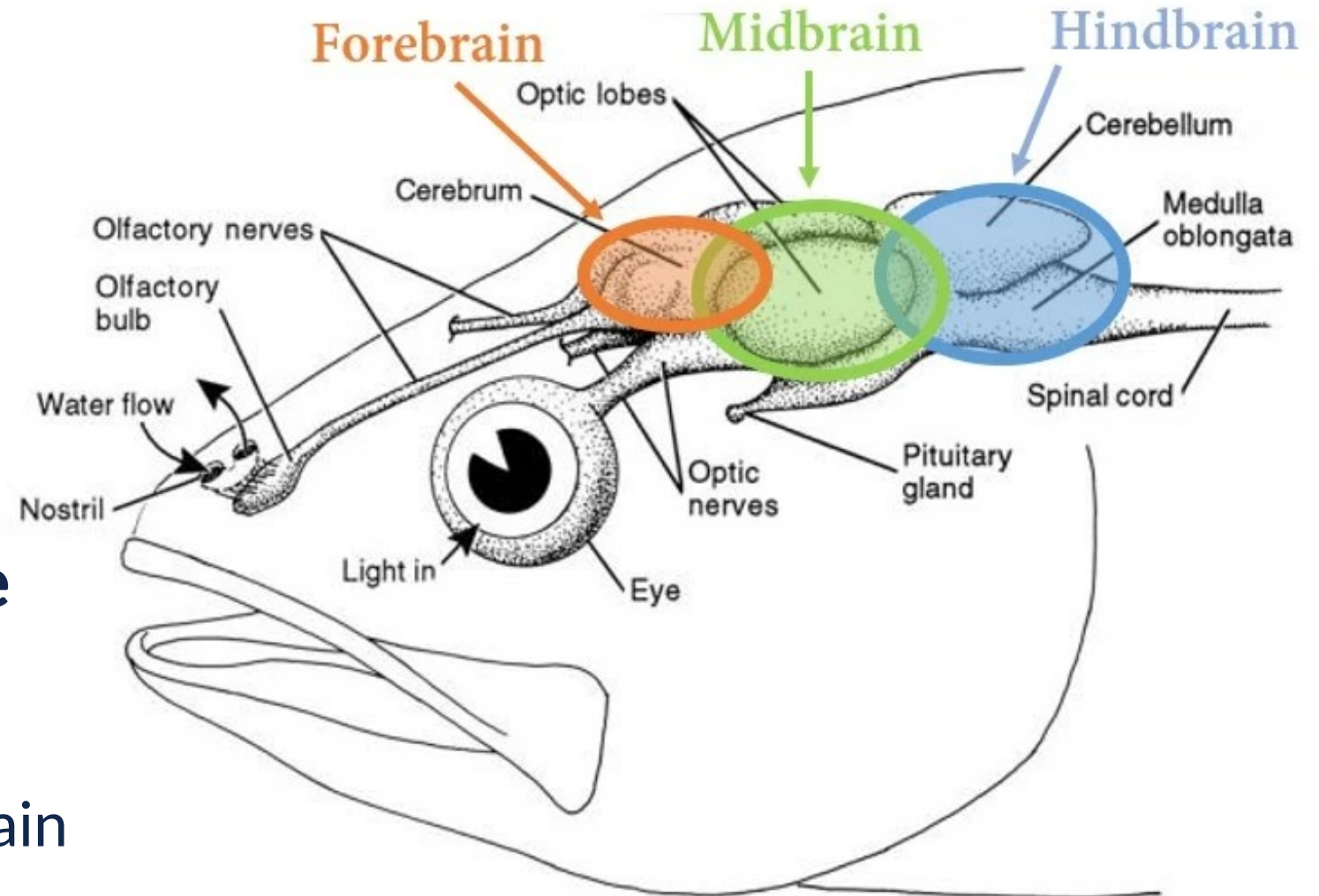
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Early Vertebrates Appear

- Big split: vertebrates vs invertebrates
- Arthropods dominate, but lack spinal cords
- Chordates: from sea squirts to fish

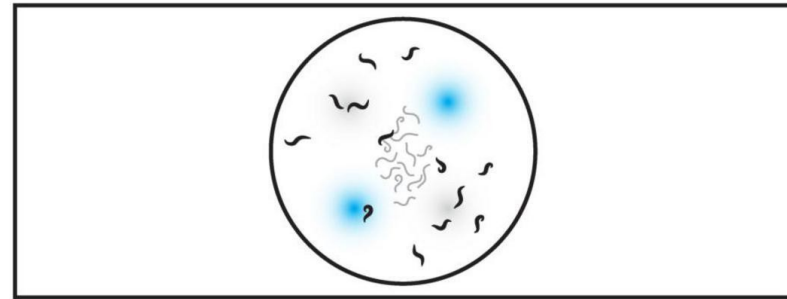
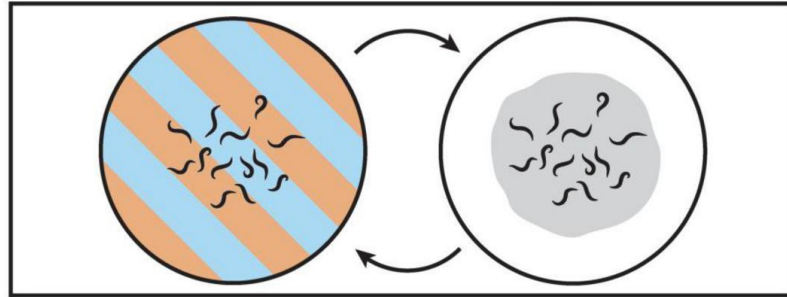
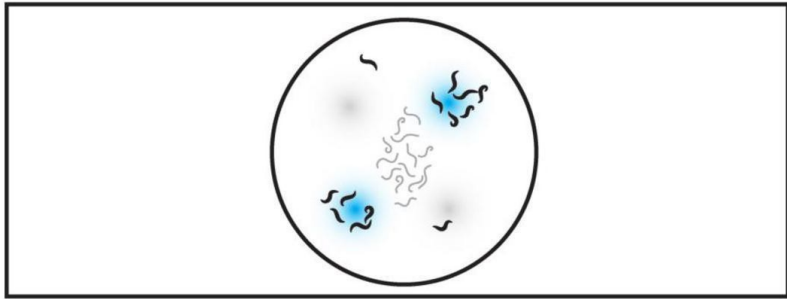
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Vertebrate Brain Template

- Shared plan across all vertebrates
- Forebrain, midbrain, hindbrain
- Basis for complex learning

Brains and Reinforcement Learning



Simple Trial-and-Error Hebb *Can* Handle

- Immediate cause and effect
- Stimuli paired closely in time
- Associations guide reflexive responses

Brains and Reinforcement Learning



Trial-and-Error With Delayed Rewards

- Actions separated from outcomes
- Sequences must be linked across time
- Simple association is not enough

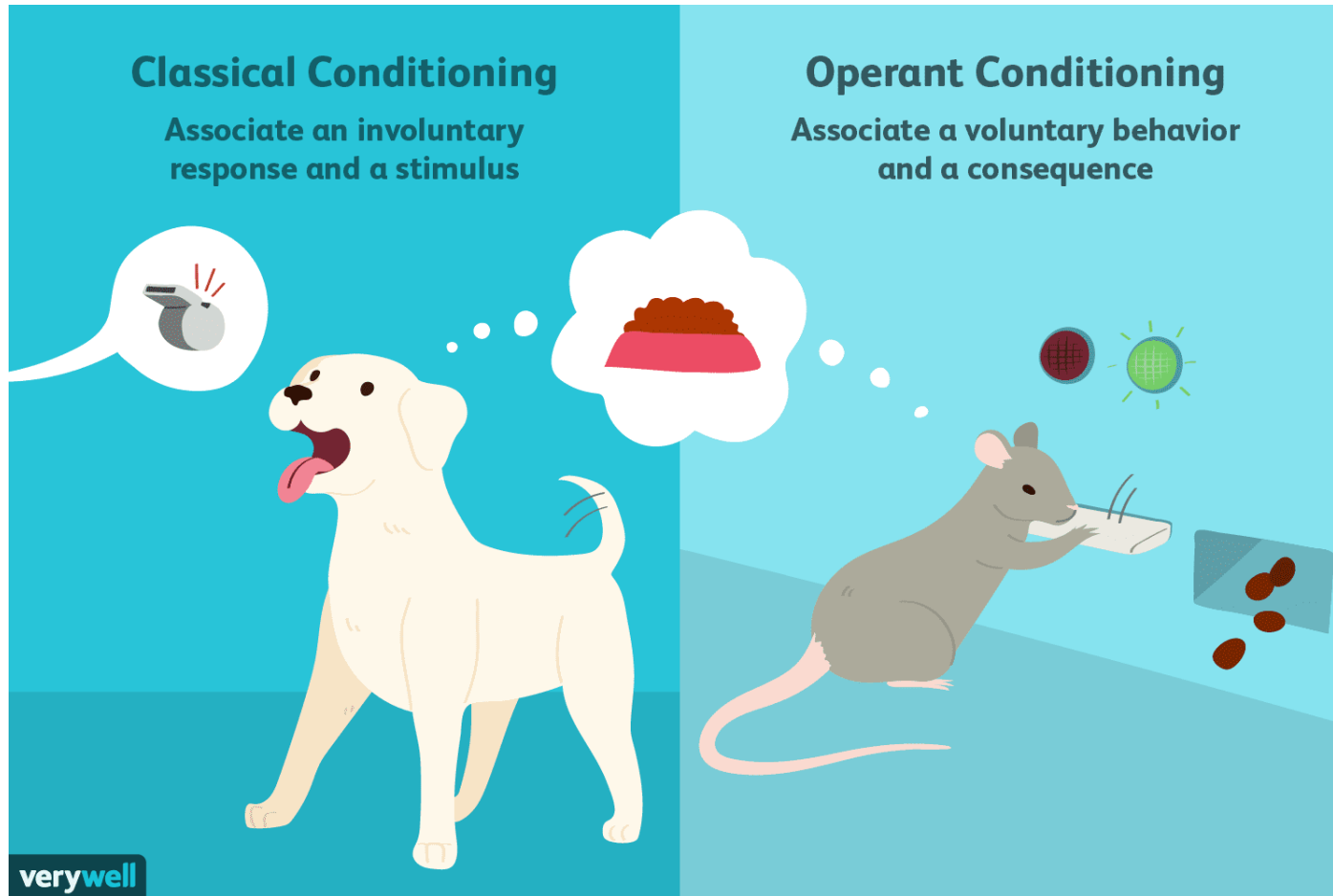
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What a Reinforcement-Learning System Must Do

- Connect actions to delayed outcomes
- Work across seconds, minutes, longer
- Balance immediate vs future rewards

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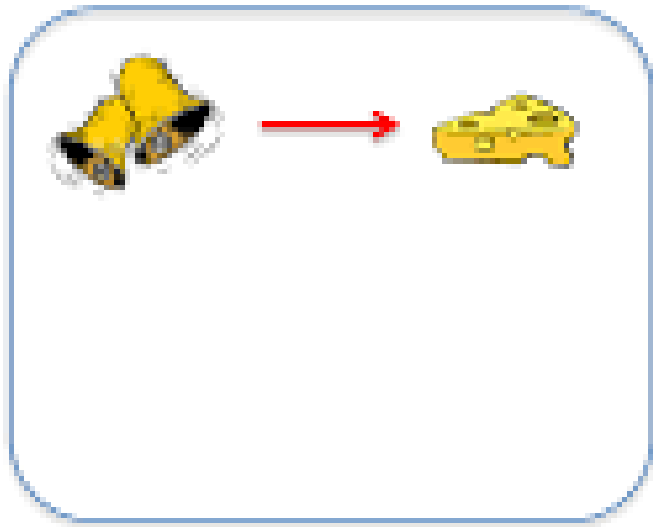


Two Families of Learning Problems

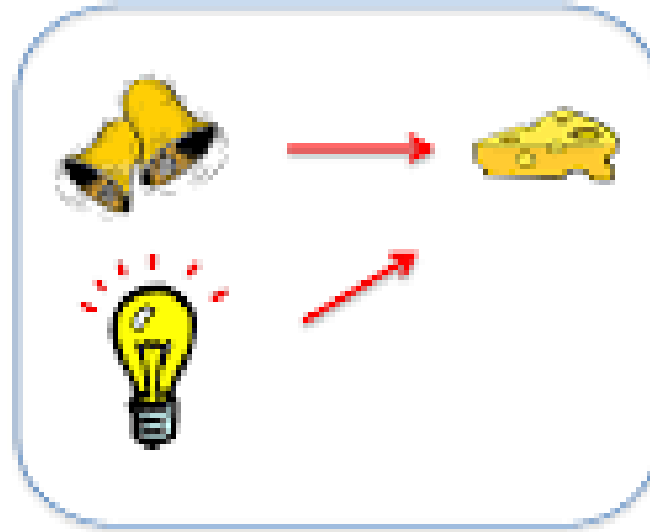
- **Classical conditioning:**
stimulus → outcome
- **Instrumental/operant conditioning:**
action → outcome
- Both shape behavior in different ways

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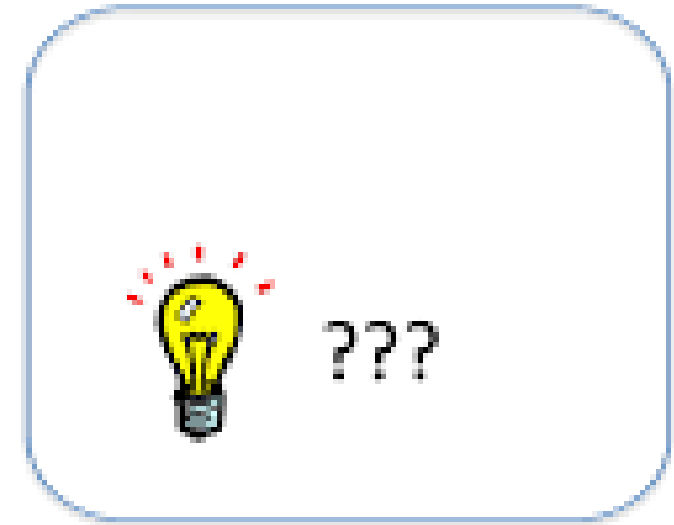
Phase 1



Phase 2



Test



Prediction Matters

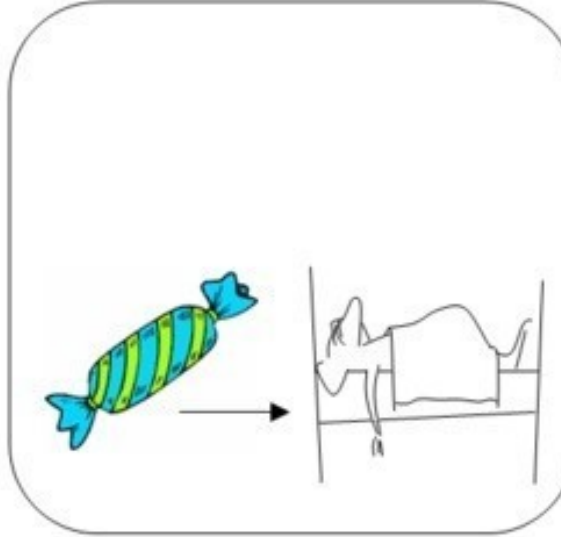
- Blocking shows limits of pairing
- Learning depends on surprise
- Prediction error drives updates

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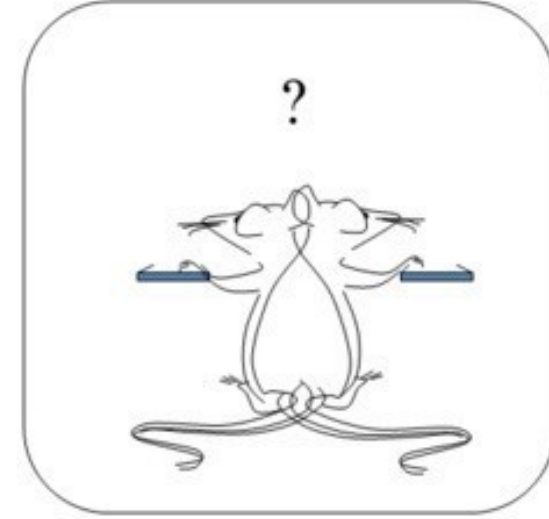
1. Instrumental Learning



2. Taste aversion learning



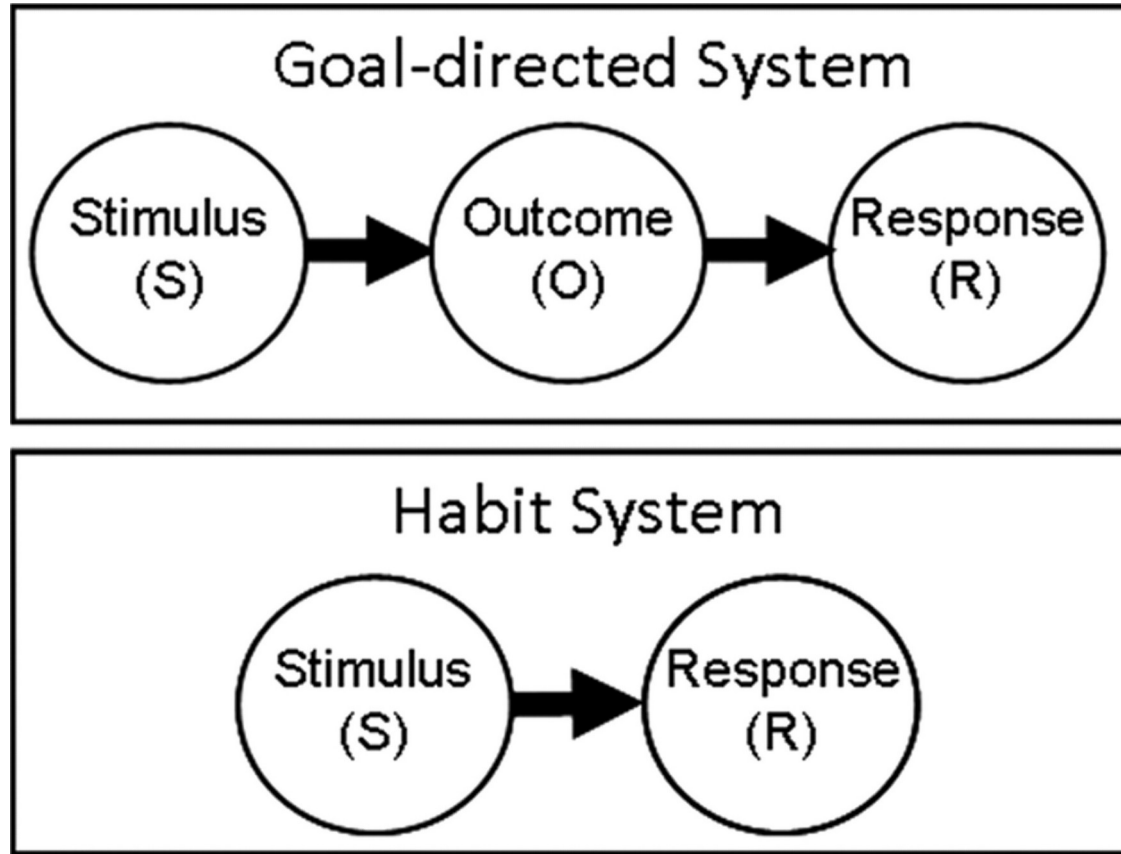
3. Test



Goals Matter

- Reinforcer devaluation
- Actions change when outcomes lose value
- Evidence for goal-directed control

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Habits vs Goal-Directed Control

- Goal-directed: sensitive to outcomes
- Habits: automatic, outcome-insensitive
- Both emerge through reinforcement

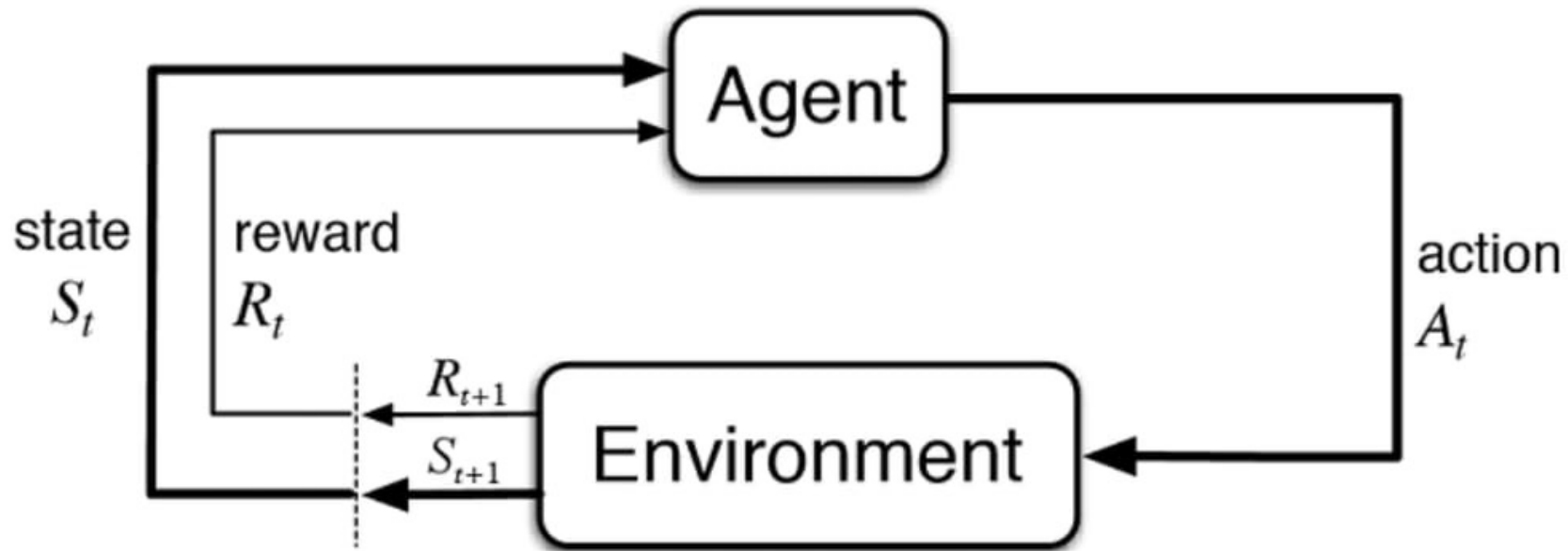
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		Expected	
		Good	Bad
Actual	Good	Satisfaction	Relief
	Bad	Disappointment	Resignation

Relief and Disappointment

- Learning from what doesn't happen
- Relief: expected bad outcome omitted
- Disappointment: expected good outcome omitted

Brains and Reinforcement Learning



Computational Definition of Reinforcement Learning

- Choose actions over time
- Maximize expected cumulative reward
- Solve credit assignment across delays

Brains and Reinforcement Learning

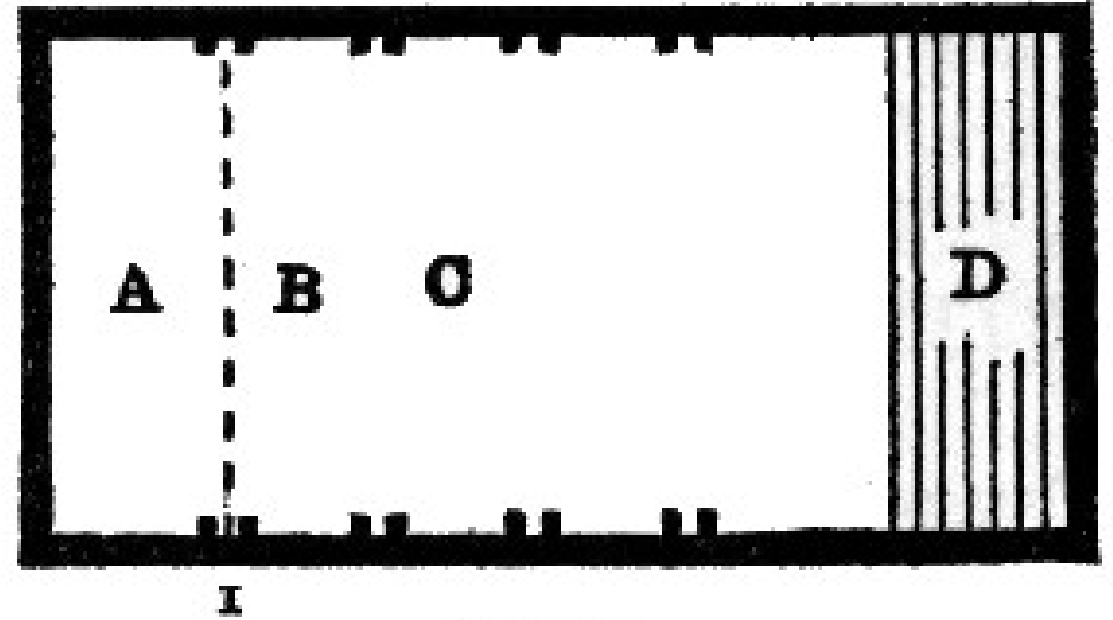


FIG. 24.

Minimal Evolutionary Requirement

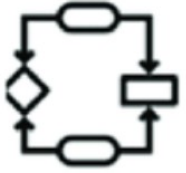
- Trial-and-error sequences
- Flexible responses, not just reflexes
- Fish succeed, worms fail

Brains and Reinforcement Learning



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

What We Learned / What's Next

- RL = actions linked to delayed outcomes
- Requires prediction error and goal sensitivity
- Next: why early AI also struggled with this